Executive Summary — Surrogate Model Validation Report

Use case: UCAIRFOILS_1 | 07 September 2026 | 1 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: MLP

Avg R^2: 0.9908 — Excellent

Q90 Accuracy — Partially met

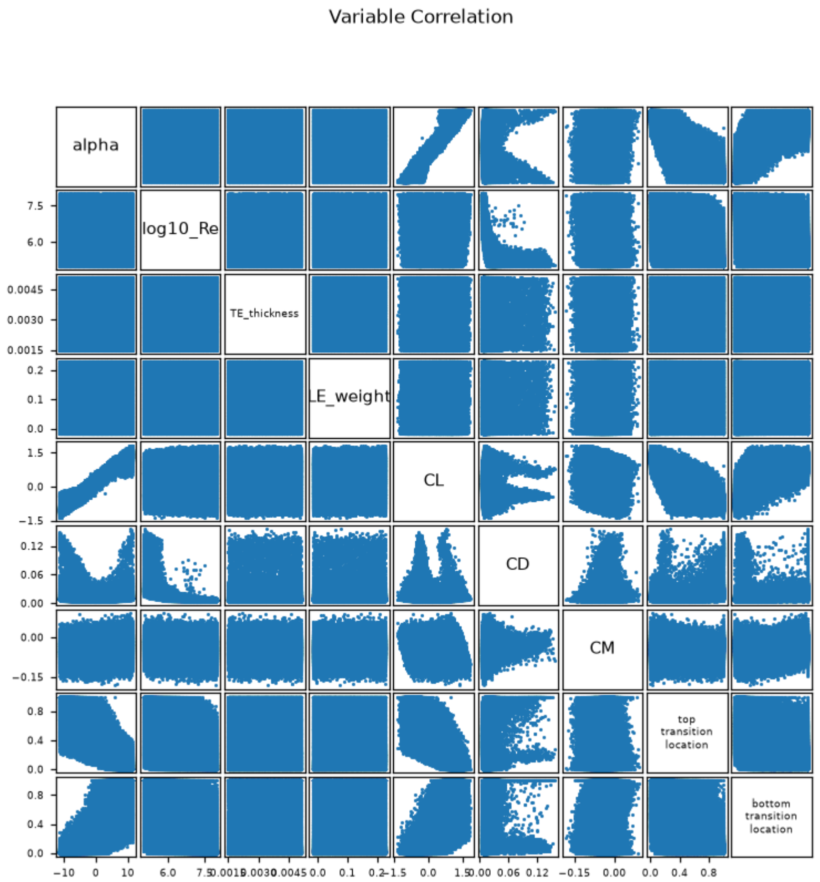
Pass: CD

Fail: CL, CM, top_transition_location, bottom_transition_location

Project Overview

Use case	UCAIRFOILS_1
Inputs	alpha, log10_Re, uW1, uW2, uW3, uW4, uW5, uW6, uW7, uW8, lW1, lW2, lW3, lW4, lW5, lW6, lW7, lW8, TE_thickness, LE_weight
Outputs	5: CL, CD, CM, top_transition_location, bottom_transition_location
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output



Data Statistics

Per-variable statistics over all data (train + val + test), as `Train_set.describe()` reports them in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum.

Inputs

	count	mean	std	min	P25	P50	P75	max
alpha	119,756	0.7531	6.3925	-11.4586	-4.5034	1.1937	6.1528	11.4591
log10_Re	119,756	6.5225	0.8611	5	5.7864	6.5347	7.2648	8
uW1	119,756	0.2303	0.0748	0.1	0.1657	0.2304	0.295	0.36
uW2	119,756	0.2299	0.075	0.1	0.1651	0.2299	0.2947	0.36
uW3	119,756	0.23	0.075	0.1	0.1651	0.2301	0.295	0.36
uW4	119,756	0.23	0.075	0.1	0.1651	0.23	0.295	0.36
uW5	119,756	0.23	0.075	0.1	0.165	0.2299	0.2949	0.36
uW6	119,756	0.23	0.075	0.1	0.1652	0.2299	0.295	0.36
uW7	119,756	0.2299	0.075	0.1	0.165	0.2299	0.2949	0.36
uW8	119,756	0.2301	0.075	0.1	0.1652	0.2302	0.2951	0.36
lW1	119,756	-0.1923	0.0877	-0.34	-0.2674	-0.1947	-0.1207	0.0872
lW2	119,756	-0.1105	0.1346	-0.34	-0.2269	-0.1128	0.0044	0.13
lW3	119,756	-0.1052	0.1354	-0.34	-0.2223	-0.1051	0.0118	0.13
lW4	119,756	-0.1045	0.1356	-0.34	-0.2218	-0.1042	0.013	0.13
lW5	119,756	-0.1048	0.1356	-0.34	-0.2222	-0.1046	0.0125	0.13
lW6	119,756	-0.1051	0.1356	-0.34	-0.2223	-0.1051	0.0123	0.13
lW7	119,756	-0.1046	0.1356	-0.34	-0.2219	-0.1043	0.0128	0.13
lW8	119,756	-0.1044	0.1355	-0.34	-0.2215	-0.104	0.0129	0.13
TE_thickness	119,756	0.0033	0.001	0.0015	0.0024	0.0033	0.0042	0.0051
LE_weight	119,756	0.1045	0.0715	-0.018	0.0425	0.1038	0.1663	0.23

Input variables — Train_set.describe()

Outputs

	count	mean	std	min	P25	P50	P75	max
CL	119,756	0.3298	0.7075	-1.3751	-0.2552	0.3931	0.9308	1.8332
CD	119,756	0.013	0.0134	0.0035	0.0068	0.0091	0.0141	0.1555
CM	119,756	-0.0444	0.0341	-0.182	-0.0676	-0.044	-0.0209	0.0925
top_transition_location	119,756	0.3748	0.2965	0	0.1243	0.2911	0.6017	1
bottom_transition_location	119,756	0.3432	0.3408	0	0.0483	0.2009	0.6041	1

Output variables — Train_set.describe()

Model Selection

Model	Avg R^2	Q90 pass (5)	KS pass (5)
MLP ⋆	0.9908	1/5	0/5

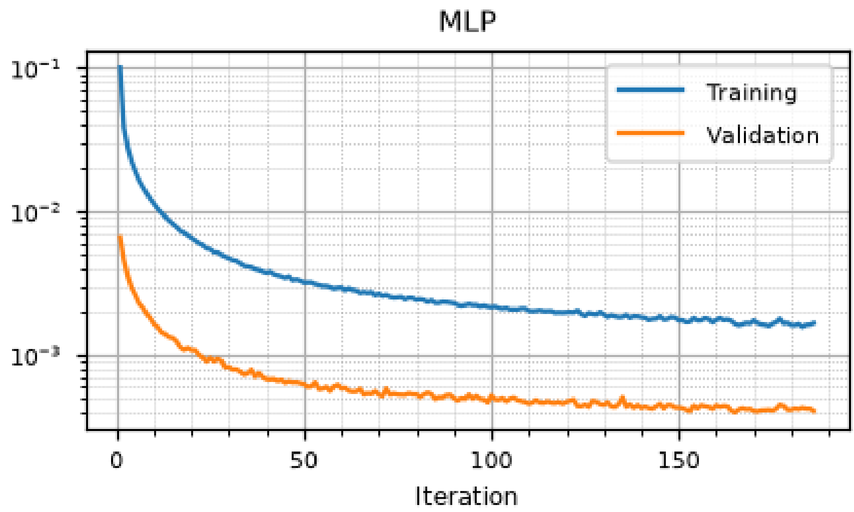
⋆ Recommended model. Q90 target: $< 10\%$. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

Output	MLP ⋆		
	R^2	Q90	Gap
CL	0.9988	0.1103	+0.0103
CD	0.9740	0.0380	-0.0620
CM	0.9919	0.2912	+0.1912
top_transition_location	0.9963	0.1520	+0.0520
bottom_transition_location	0.9931	0.2881	+0.1881

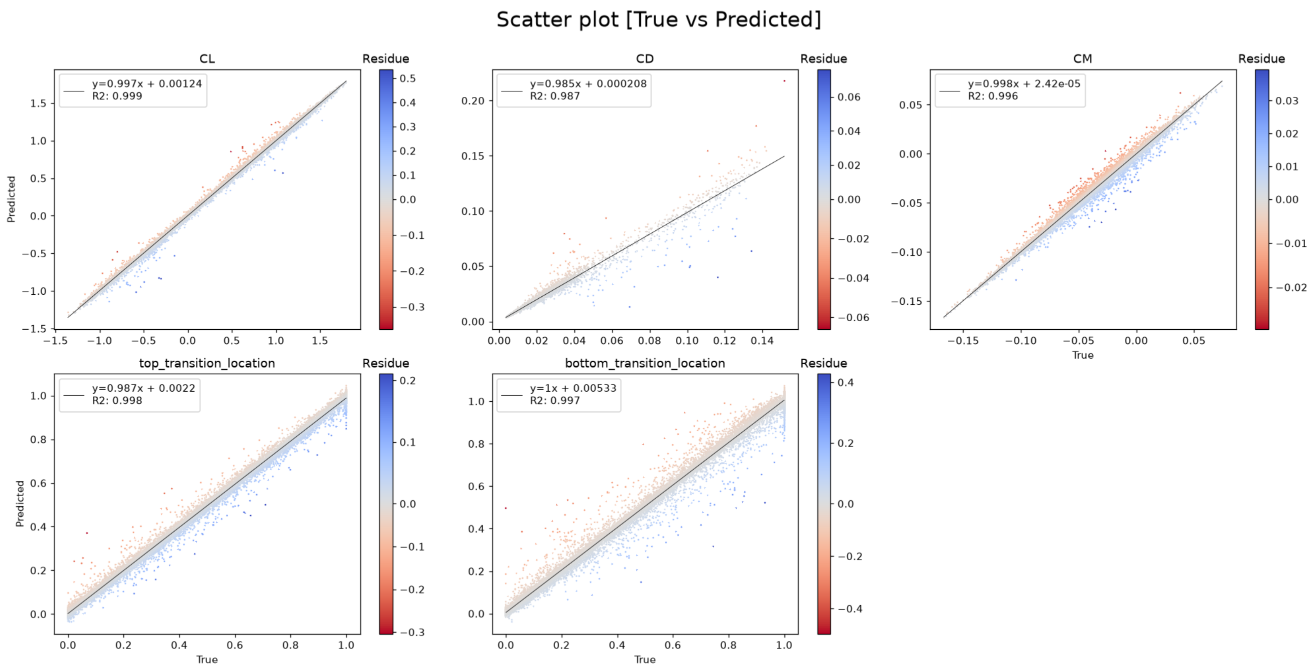
$R^2 \geq 0.95$ $R^2 \in [0.80, 0.95)$ $R^2 < 0.80$ or Q90 fail pass Gap=Q90−target ($<0 \Rightarrow$ margin)

Training History



Training and validation loss per iteration, log scale. For models trained with early stopping, the validation loss is approximated from the recorded validation R^2 via $(1 - R^2) \cdot \mathrm{Var}(y)/2$. Gradient boosting reports per-stage deviance averaged over its per-output estimators.

Predicted vs True — MLP



Data Split Quality

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.965	
p-hacking test proportion (≤ 0.05 warn)	0.003	✓
Isolated test proportion (≤ 0.05 warn)	0.033	✓
Chi^2 p-value (≥ 0.05)	—	—

No p-hacking concern: only 0.2% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Overfitting Check (KS Test) — MLP

Output	KS p-value
CL	0.000
CD	0.000
CM	0.000
top_transition_location	0.000
bottom_transition_location	0.002

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Deep Analysis

UCAIRFOILS_1

Winner Model — Full Validation

Deep Analysis — MLP

Mirrors `validation_template.ipynb` — all computations from validation CSVs

Data Overview

Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

Input Statistics

	count	mean	std	min	P25	P50	P75	max
alpha	119,756	0.7531	6.3925	-11.4586	-4.5034	1.1937	6.1528	11.4591
log10_Re	119,756	6.5225	0.8611	5	5.7864	6.5347	7.2648	8
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uW3	119,756	0.23	0.075	0.1	0.1651	0.2301	0.295	0.36
uW4	119,756	0.23	0.075	0.1	0.1651	0.23	0.295	0.36
uW5	119,756	0.23	0.075	0.1	0.165	0.2299	0.2949	0.36
uW6	119,756	0.23	0.075	0.1	0.1652	0.2299	0.295	0.36
uW7	119,756	0.2299	0.075	0.1	0.165	0.2299	0.2949	0.36
uW8	119,756	0.2301	0.075	0.1	0.1652	0.2302	0.2951	0.36
lW1	119,756	-0.1923	0.0877	-0.34	-0.2674	-0.1947	-0.1207	0.0872
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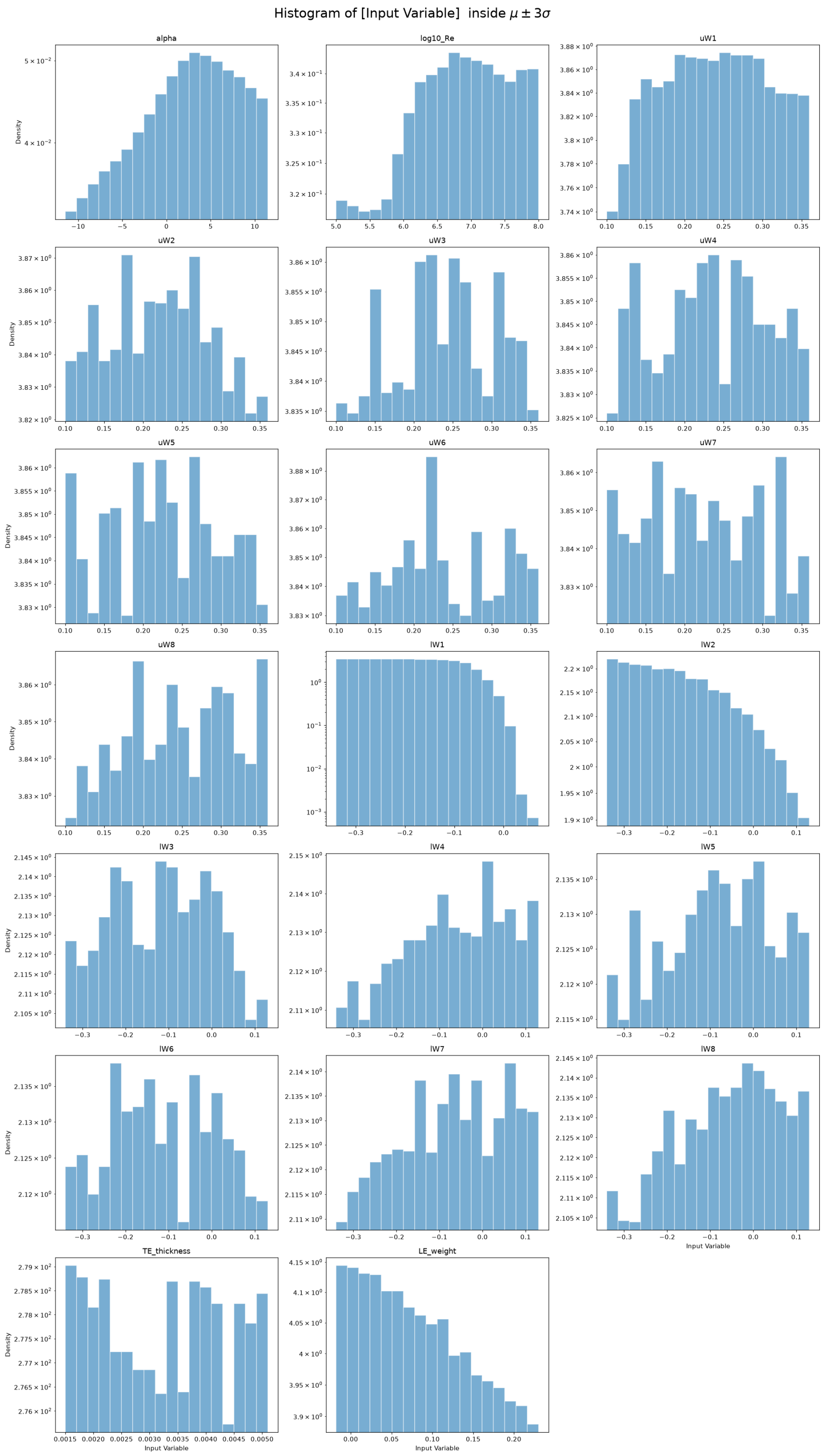
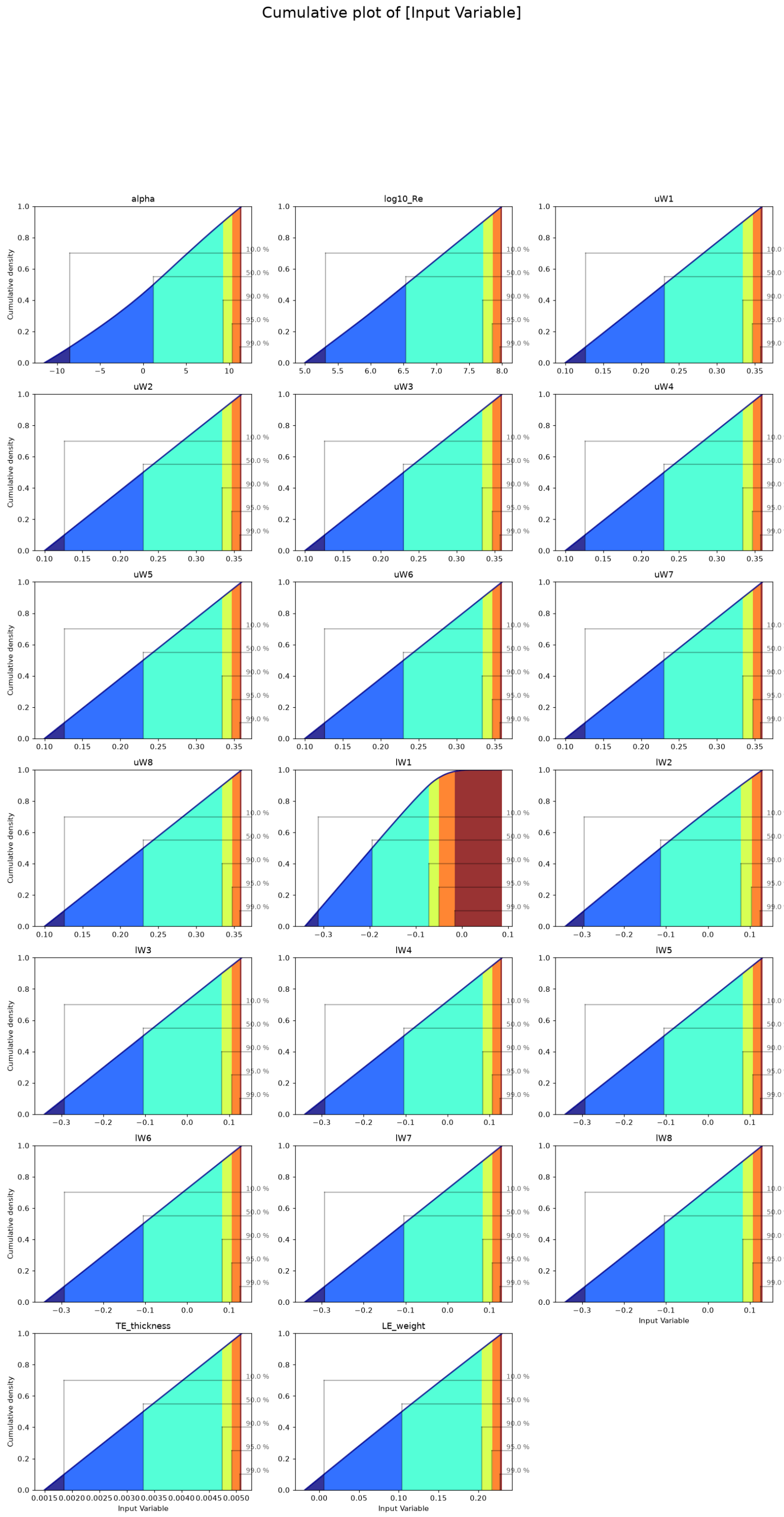
TE_thickness	119,756	0.0033	0.001	0.0015	0.0024	0.0033	0.0042	0.0051
LE_weight	119,756	0.1045	0.0715	-0.018	0.0425	0.1038	0.1663	0.23

Input variables — Train_set.describe()

Input Histograms



Input Cumulative Distributions

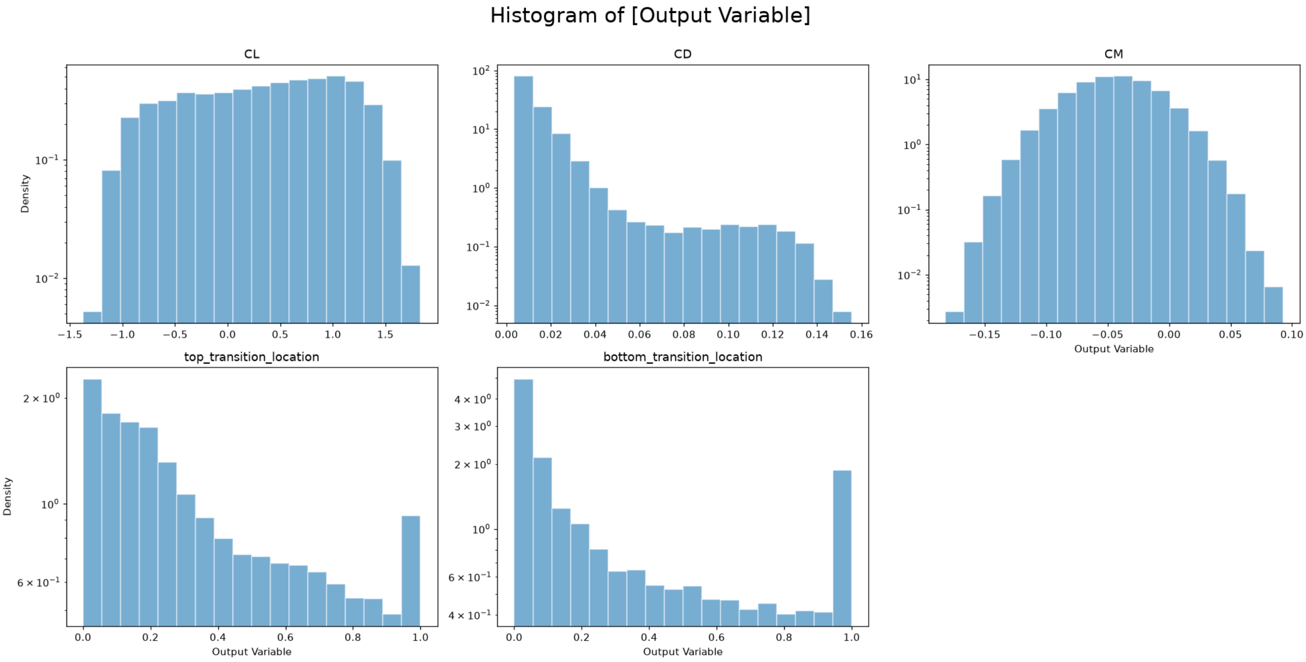


Output Statistics

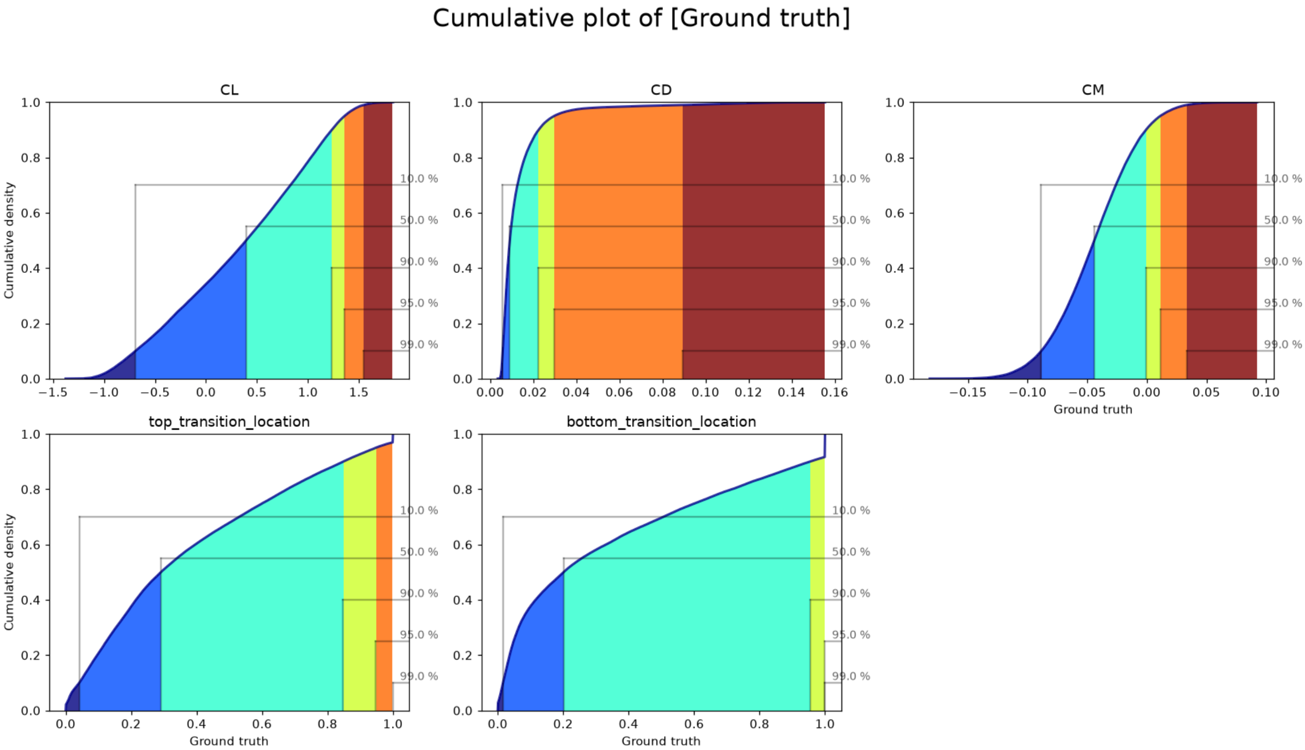
	count	mean	std	min	P25	P50	P75	max
CL	119,756	0.3298	0.7075	-1.3751	-0.2552	0.3931	0.9308	1.8332
CD	119,756	0.013	0.0134	0.0035	0.0068	0.0091	0.0141	0.1555
CM	119,756	-0.0444	0.0341	-0.182	-0.0676	-0.044	-0.0209	0.0925
top_transition_location	119,756	0.3748	0.2965	0	0.1243	0.2911	0.6017	1
bottom_transition_location	119,756	0.3432	0.3408	0	0.0483	0.2009	0.6041	1

Output variables — Train_set.describe()

Output Histograms



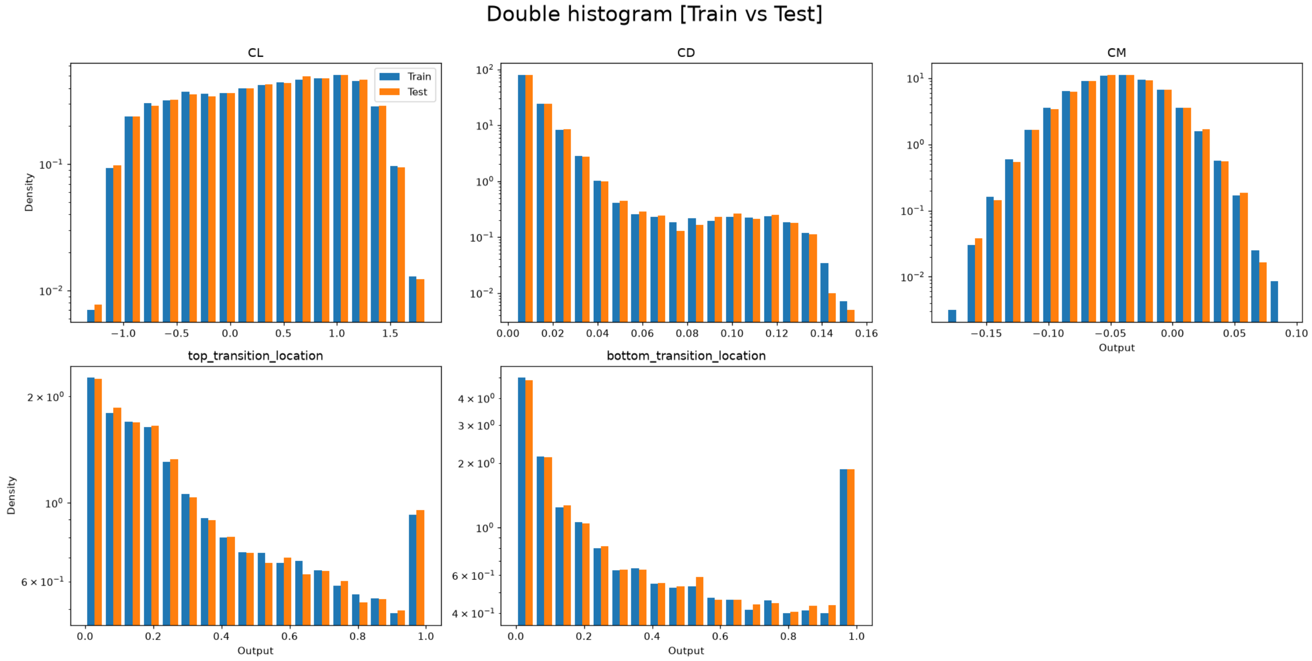
Output Cumulative Distributions



Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0: both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test

KS and AD Test Results

	KS	AD
CL	0.0905	0.1513
CD	0.3411	0.25
CM	0.3781	0.25
top_transition_location	0.5412	0.25
bottom_transition_location	0.0754	0.0371

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Split Quality — Voxel Tesselation Proximity (p-hacking check)

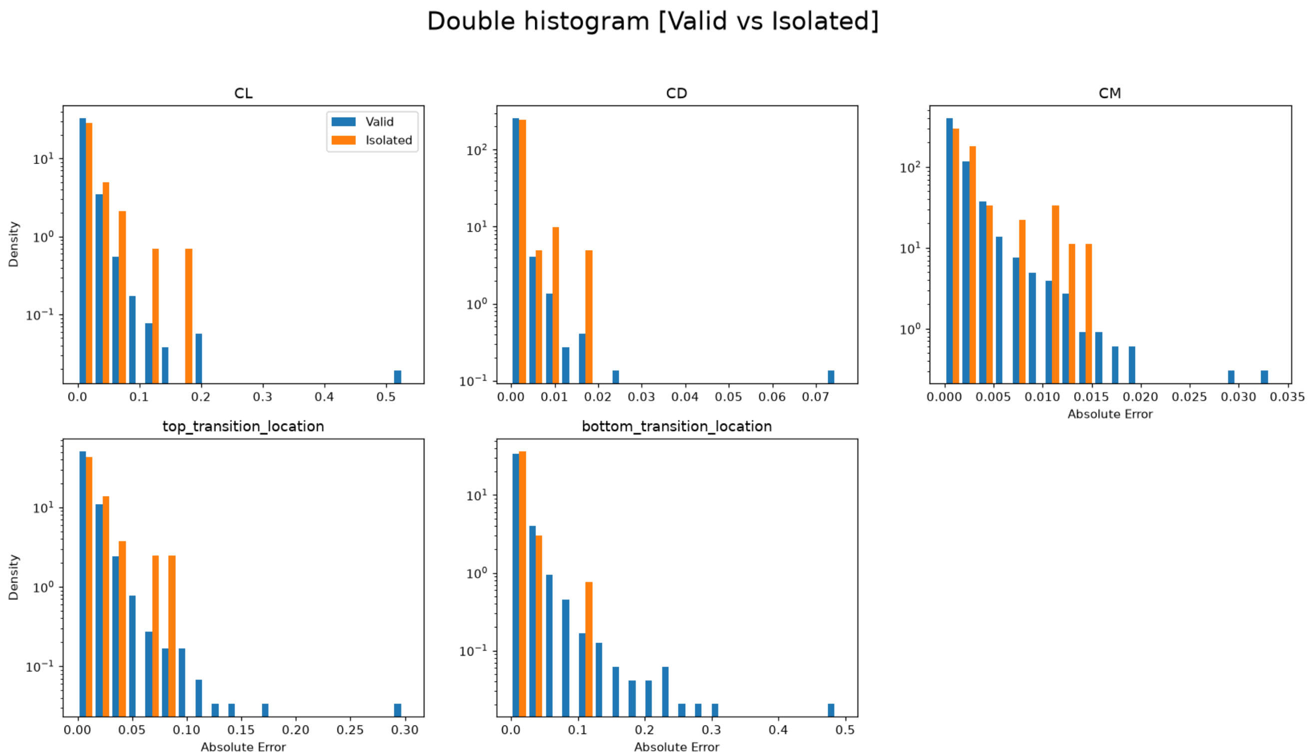
Beyond matching distributions, a split can still flatter the model if test points sit right next to training points (p-hacking) or probe regions the model never saw (isolated, residual voxel). The VTPM method from `validationlib` (SF_9 cell 9.0) classifies every test point.

Metric	Value	Pass
Residual voxel proportion ($\leq$0.05)	0.000	✓
Valid test proportion	0.965	
p-hacking test proportion ($\leq$0.05 warn)	0.003	✓
Isolated test proportion ($\leq$0.05 warn)	0.033	✓
Chi^2 p-value ($\geq$0.05)	—	—

No p-hacking concern: only 0.2% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Error Distribution: Valid vs Isolated Test Points

Same comparison for the test points with no training neighbour.



	AD	KS	KW
CL	0.0256	0.0581	0.0358
CD	0.0034	0.0327	0.0142
CM	0.0156	0.0111	0.0405
top_transition_location	0.1573	0.3304	0.2561
bottom_transition_location	0.25	0.657	0.3696

Valid ($n = 1946$) vs Isolated ($n = 53$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
CL	23,952	-0.363	0.537	-0.000348	$^{-0.000103}_{-0.000672}$	-0.0012	$^{-0.001}_{-0.0015}$	0.0249	$^{0.0261}_{0.024}$	0.0213	$^{0.0216}_{0.0209}$	54.4	$^{73.6}_{34.4}$	2.21	$^{3.41}_{0.935}$
CD	23,952	-0.0661	0.0762	-1.37e-05	$^{1.49e-05}_{-4.62e-05}$	-6.26e-06	$^{-3.26e-06}_{-9.54e-06}$	0.0022	$^{0.0024}_{0.0019}$	0.000346	$^{0.000351}_{0.00034}$	307	$^{403}_{202}$	5.98	$^{10.6}_{-0.876}$
CM	23,952	-0.0296	0.0395	-0.000119	$^{-7.4e-05}_{-0.00016}$	-0.000183	$^{-0.000153}_{-0.000202}$	0.0031	$^{0.0031}_{0.003}$	0.0022	$^{0.0022}_{0.0022}$	17	$^{19.9}_{14.3}$	1.04	$^{1.38}_{0.68}$
top_transition_location	23,952	-0.304	0.211	0.0028	$^{0.003}_{0.0025}$	0.0018	$^{0.0019}_{0.0016}$	0.0178	$^{0.0184}_{0.0174}$	0.0152	$^{0.0154}_{0.015}$	20.3	$^{27.5}_{15.2}$	0.0865	$^{0.581}_{-0.641}$
bottom_transition_location	23,952	-0.496	0.428	-0.0053	$^{-0.005}_{-0.0057}$	-0.0053	$^{-0.0052}_{-0.0054}$	0.0278	$^{0.0288}_{0.0267}$	0.016	$^{0.0162}_{0.0157}$	33.7	$^{39.6}_{26.3}$	0.483	$^{1.29}_{-0.241}$

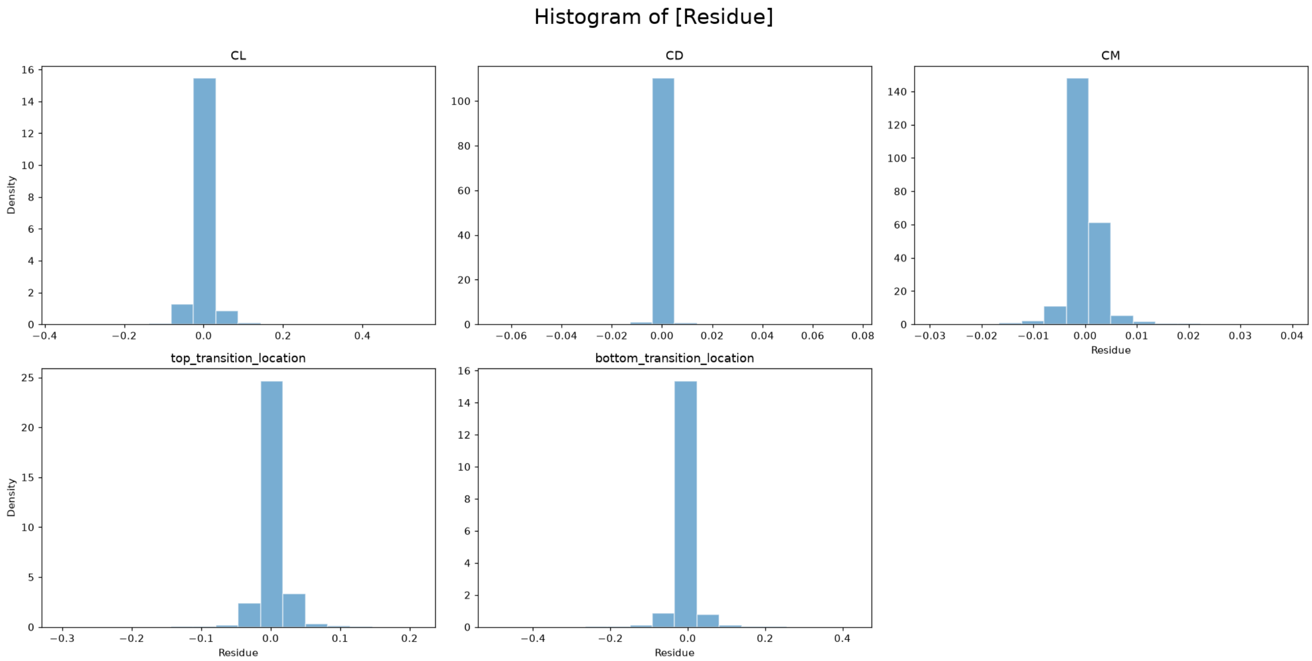
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

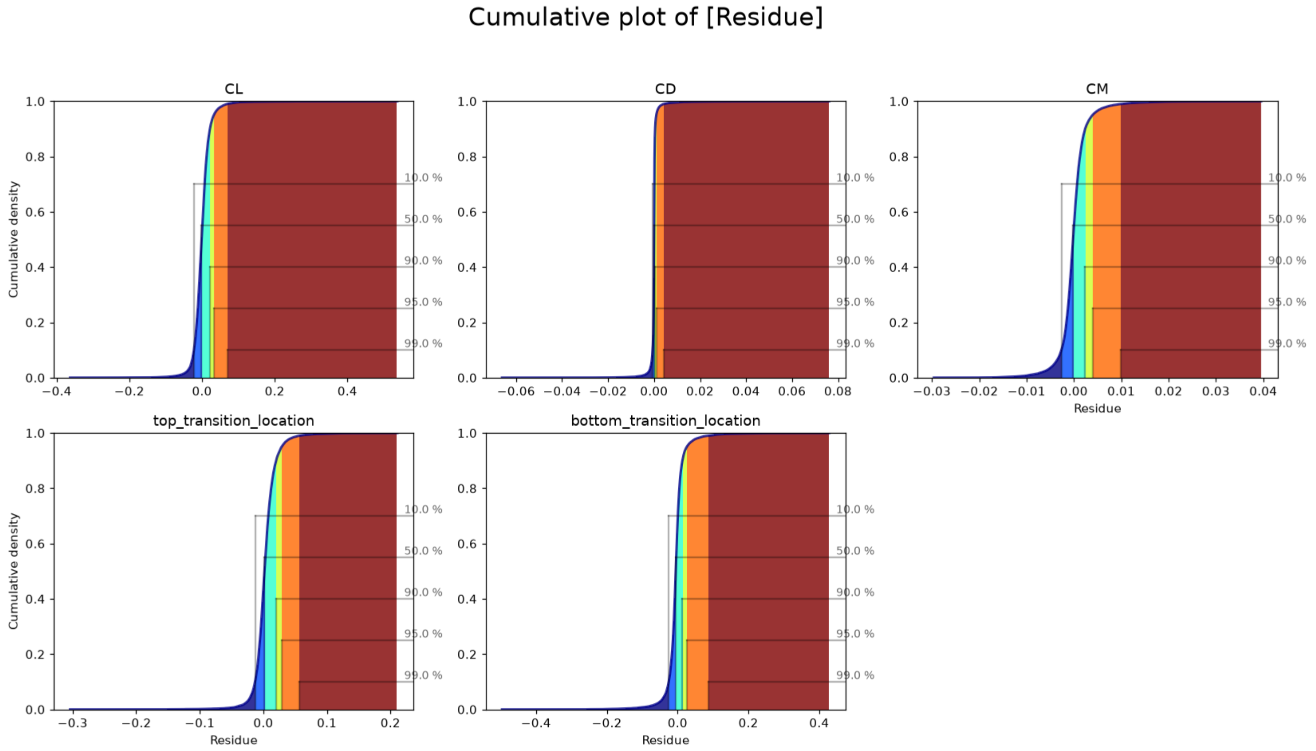
	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
CL	23,952	1.28e-06	0.537	0.0153	$^{0.0155}_{0.015}$	0.0106	$^{0.0108}_{0.0104}$	0.0196	$^{0.0209}_{0.0184}$	0.0143	$^{0.0145}_{0.0141}$	118	$^{153}_{74.3}$	7.55	$^{8.81}_{6.09}$
CD	23,952	1.09e-08	0.0762	0.000577	$^{0.000601}_{0.000556}$	0.00017	$^{0.000173}_{0.000166}$	0.0021	$^{0.0023}_{0.0019}$	0.000343	$^{0.000352}_{0.000337}$	339	$^{447}_{239}$	15.1	$^{16.9}_{12.9}$
CM	23,952	1.46e-07	0.0395	0.0018	$^{0.0019}_{0.0018}$	0.0011	$^{0.0011}_{0.0011}$	0.0024	$^{0.0025}_{0.0023}$	0.0016	$^{0.0017}_{0.0016}$	29	$^{36.7}_{23.4}$	4.23	$^{4.63}_{3.92}$
top_transition_location	23,952	1.01e-06	0.304	0.0114	$^{0.0115}_{0.0113}$	0.0076	$^{0.0077}_{0.0075}$	0.0139	$^{0.0144}_{0.0135}$	0.011	$^{0.0111}_{0.0107}$	39.9	$^{54.2}_{28.2}$	4.61	$^{5.24}_{4.04}$
bottom_transition_location	23,952	1.47e-07	0.496	0.0157	$^{0.016}_{0.0153}$	0.0093	$^{0.0094}_{0.0091}$	0.0236	$^{0.0245}_{0.0224}$	0.0136	$^{0.0139}_{0.0133}$	50.1	$^{60.5}_{38.5}$	5.58	$^{6.06}_{5.11}$

Absolute-error statistics with bootstrap confidence intervals.

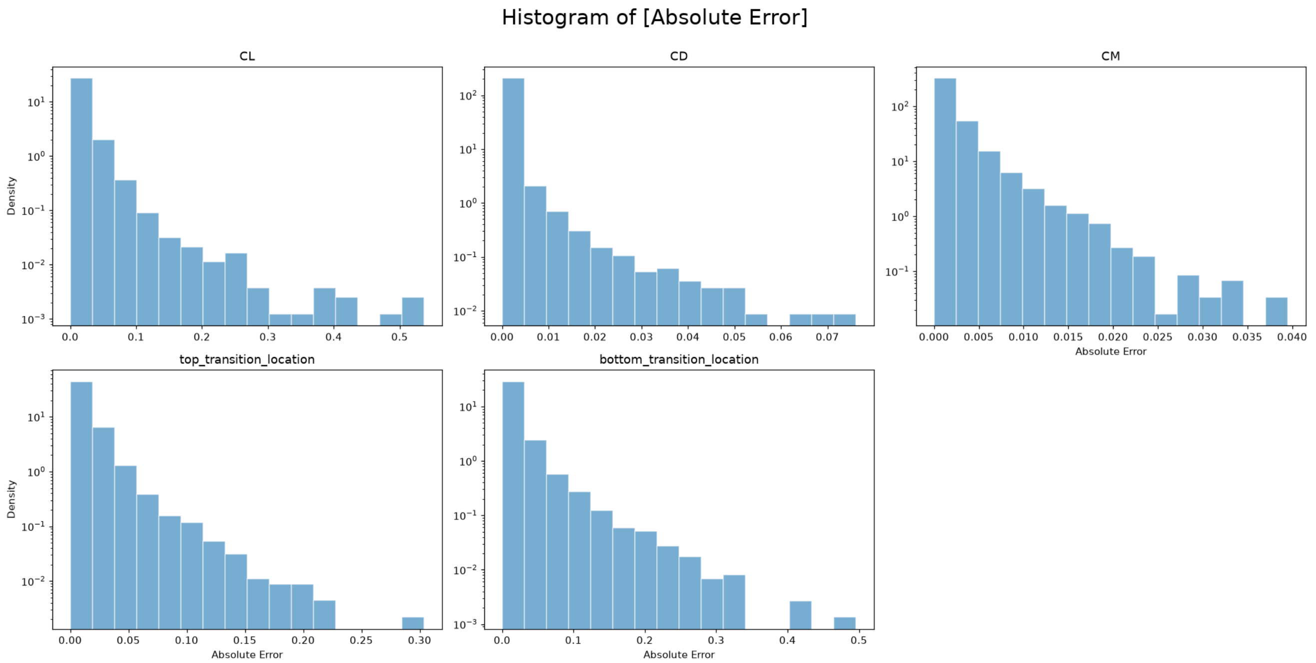
Residue Histograms



Residue CDF



Absolute Error Histograms



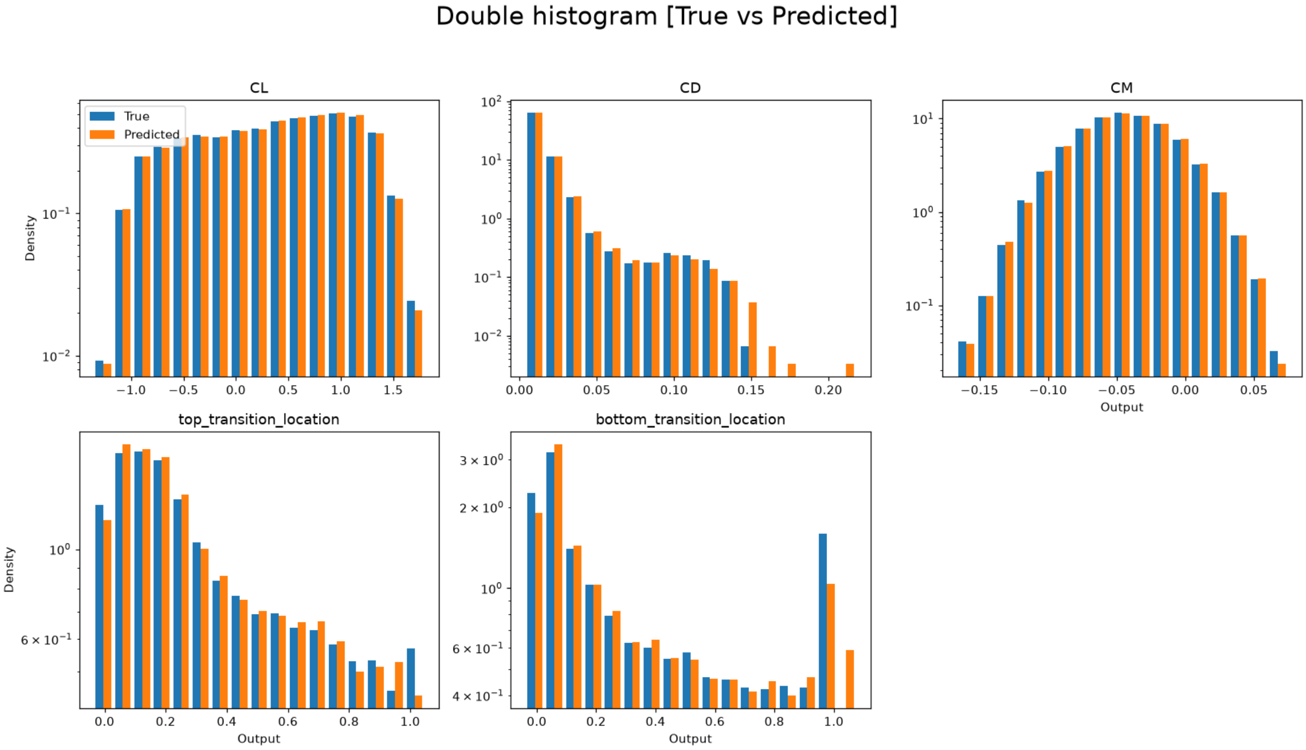
Q90 Accuracy Requirement

Output	Q90	Target	Status
CL	0.1107	10%	FAIL
CD	0.0632	10%	PASS
CM	0.2174	10%	FAIL
top_transition_location	0.1554	10%	FAIL
bottom_transition_location	0.3584	10%	FAIL

$p \geq 0.05$ — pass $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

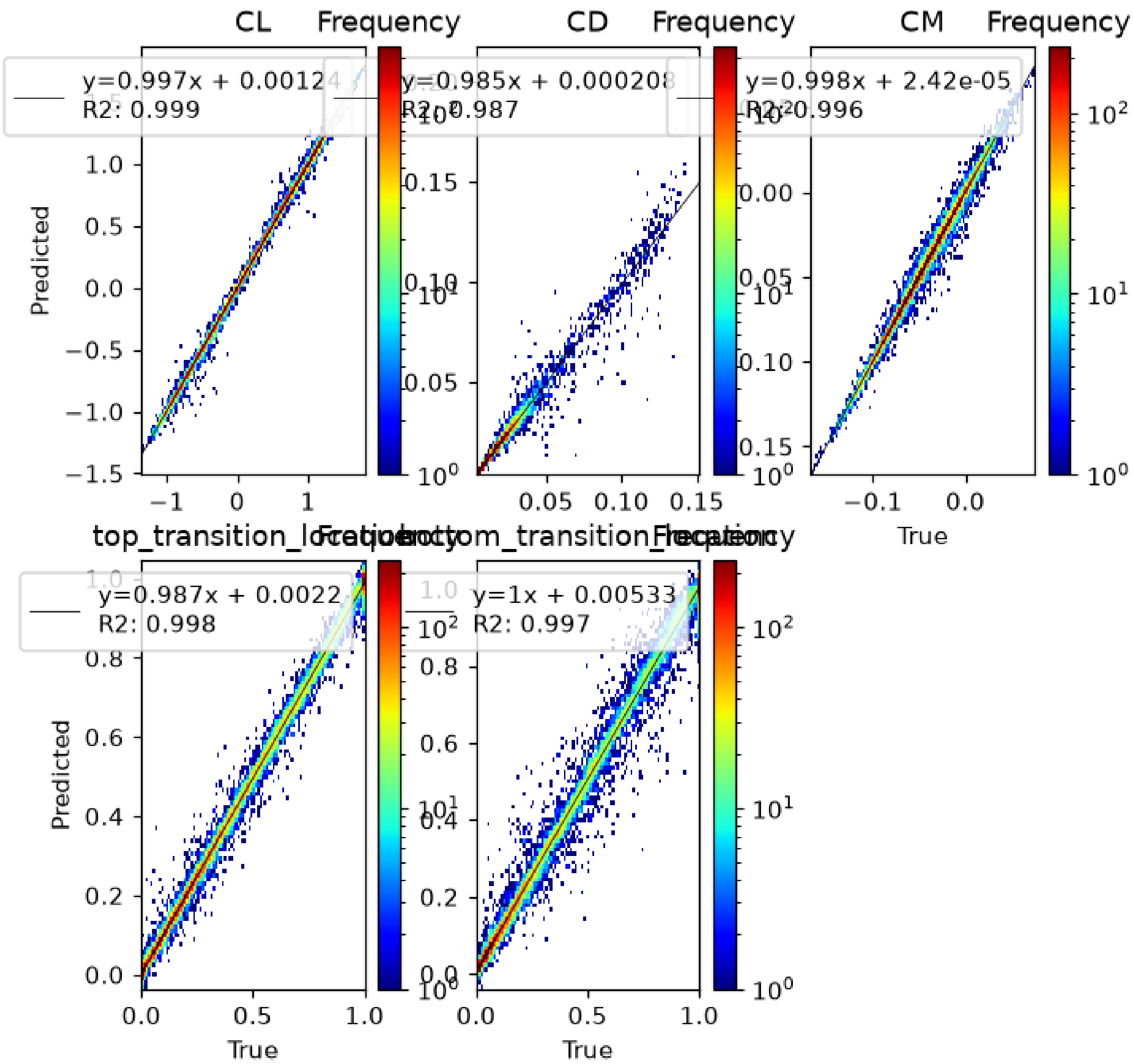


	AD	KS
CL	0.25	0.9942
CD	0.25	0.9957
CM	0.25	0.9671
top_transition_location	0.0014	0.000816
bottom_transition_location	0.001	3.9e-37

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.



P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: alpha→CL, alpha→CD, alpha→CM, alpha→top_transition_locatio alpha→bottom_transition_location, log10_Re→CL, log10_Re→CD, log10_Re→CM, log10_Re→top_transition_location, log10_Re→bottom_transition_location.... These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

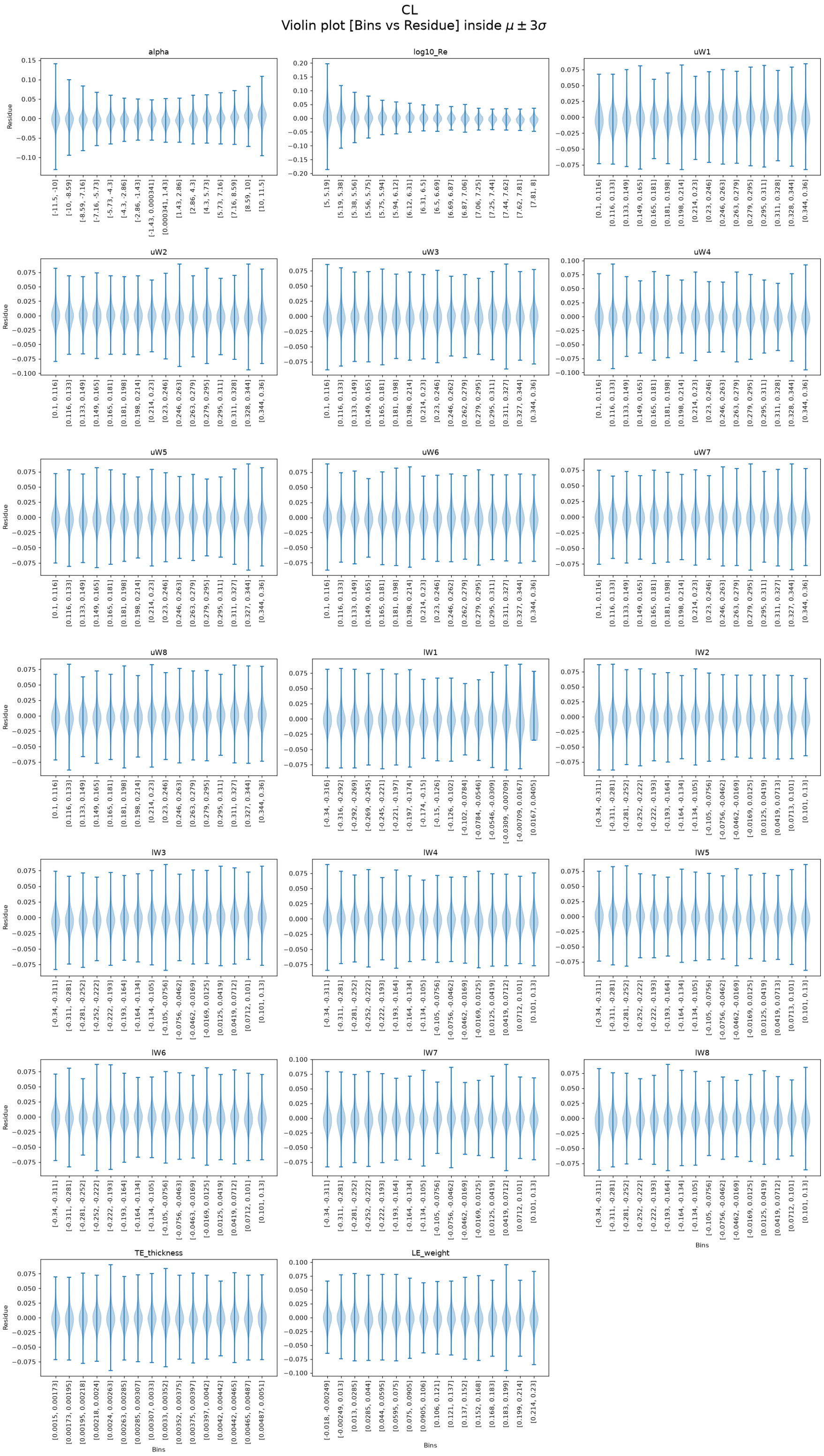
Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

	CL	CD	CM	top_transition_location	bottom_transition_location
alpha_binned	7.21e-205	1.32e-18	2.59e-38	0	1.05e-108
log10_Re_binned	8.67e-158	2.76e-82	1.56e-06	9.39e-41	1.06e-10
uW1_binned	1.5e-28	0.0843	4.28e-19	0.1011	5.84e-05
uW2_binned	4.71e-17	0.6386	1.05e-05	0.2958	0.6552
uW3_binned	5.84e-05	0.9046	1.01e-08	0.0234	0.573
uW4_binned	0.3469	0.8533	0.802	0.224	0.1802
uW5_binned	0.000253	0.0016	1.16e-05	6.54e-18	0.0201
uW6_binned	0.0275	0.3202	0.0365	2.98e-07	0.000288
uW7_binned	0.7438	0.0578	0.0831	0.0053	0.629
uW8_binned	9.83e-40	0.0015	8.89e-05	0.0416	0.0041
lW1_binned	0.0171	9.81e-07	4.45e-07	3.05e-09	1.17e-33
lW2_binned	0.0013	0.2272	2.89e-09	0.0681	0.9189
lW3_binned	5.64e-73	0.1178	0.0332	0.0793	2.93e-12
lW4_binned	2.38e-21	0.1198	0.1955	7.97e-12	0.014
lW5_binned	5.35e-17	1.61e-07	0.009	0.002	3.79e-14
lW6_binned	0.5053	0.0056	0.036	0.106	7.13e-06
lW7_binned	5.25e-10	0.000141	0.0279	0.0859	0.0039
lW8_binned	2.43e-13	0.0571	0.00044	0.1833	0.000288
TE_thickness_binned	0.0767	6.57e-22	1.06e-14	0.2361	0.6597
LE_weight_binned	0.0023	0.3054	0.0595	0.0169	0.2745

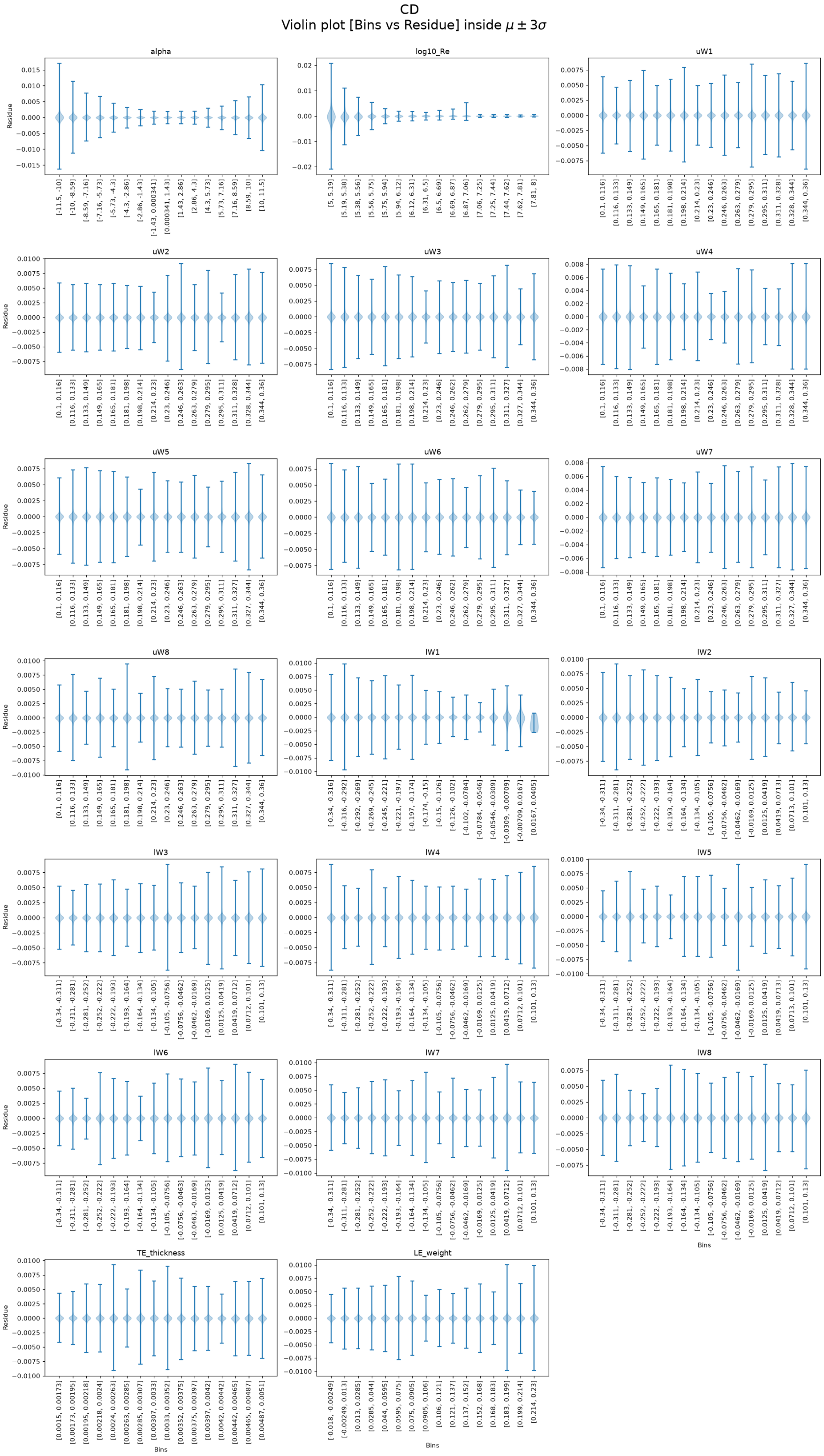
Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

Residue Violin Plots per Output (binned by input)

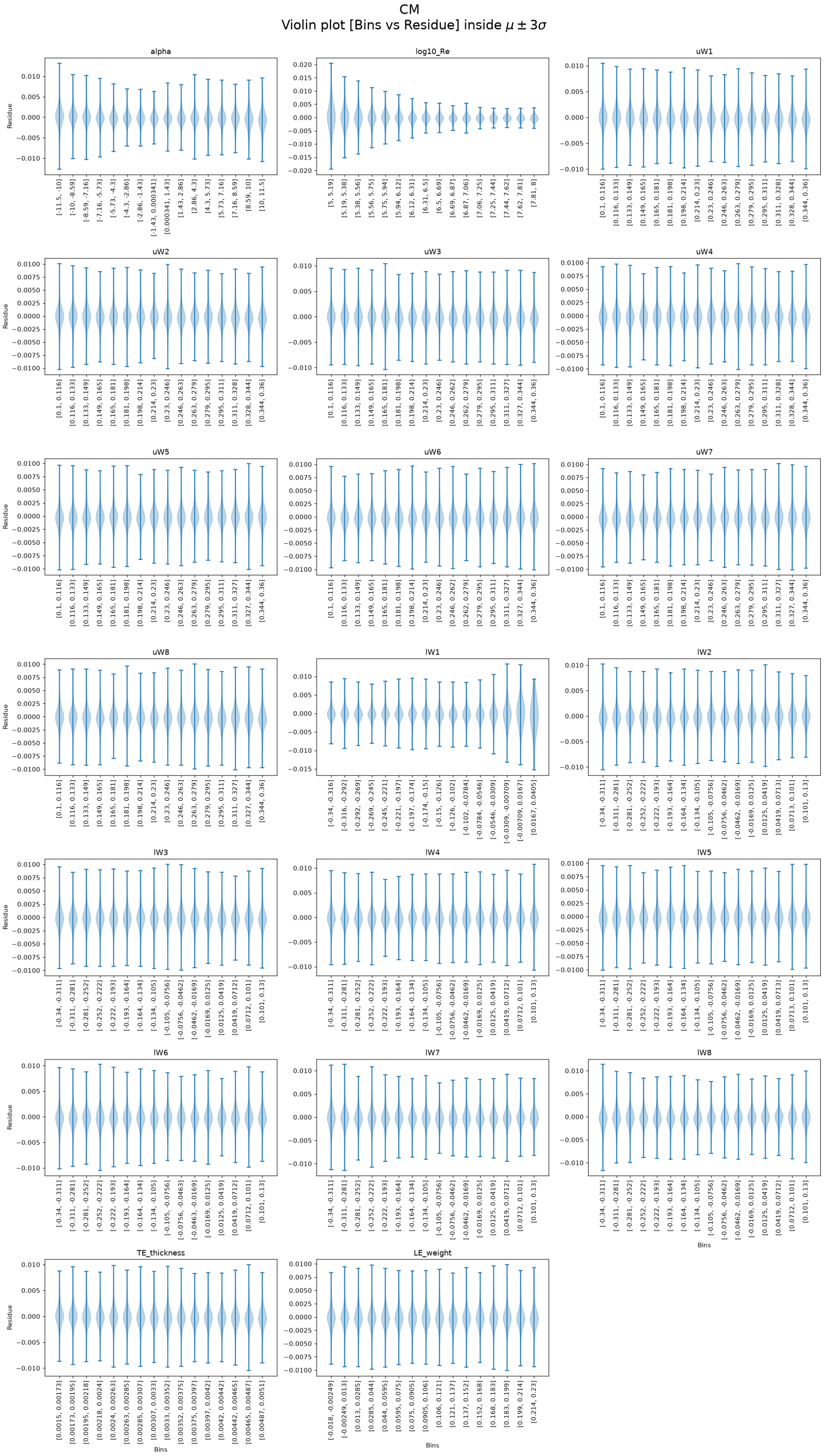
Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.
Residue — CL



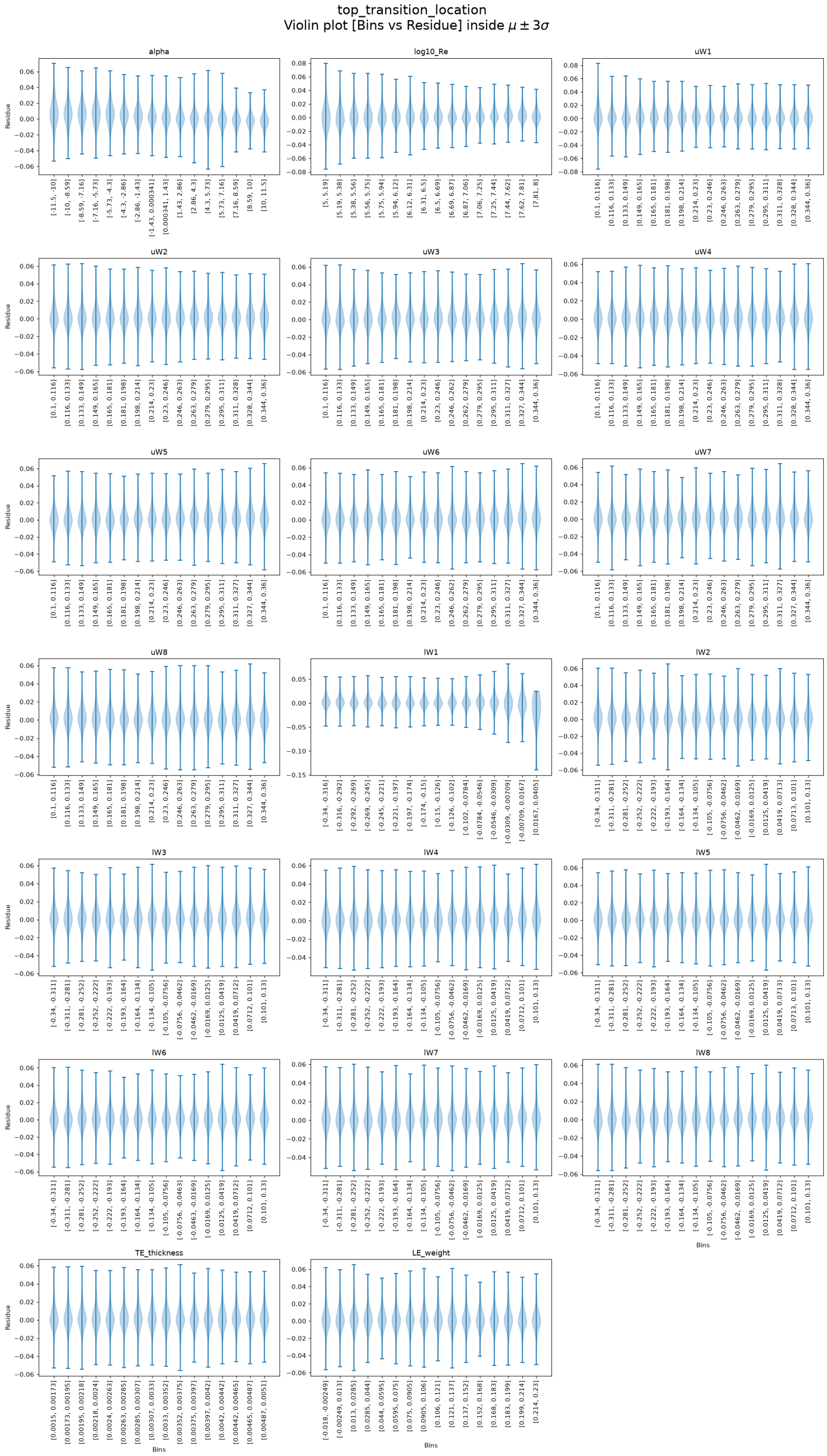
Residue — CD


CD
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$
alpha
log10_Re
uW1
uW2
uW3
uW4
uW5
uW6
uW7
uW8
lW1
lW2
lW3
lW4
lW5
lW6
lW7
lW8
TE_thickness
LE_weight
Residue
Bins

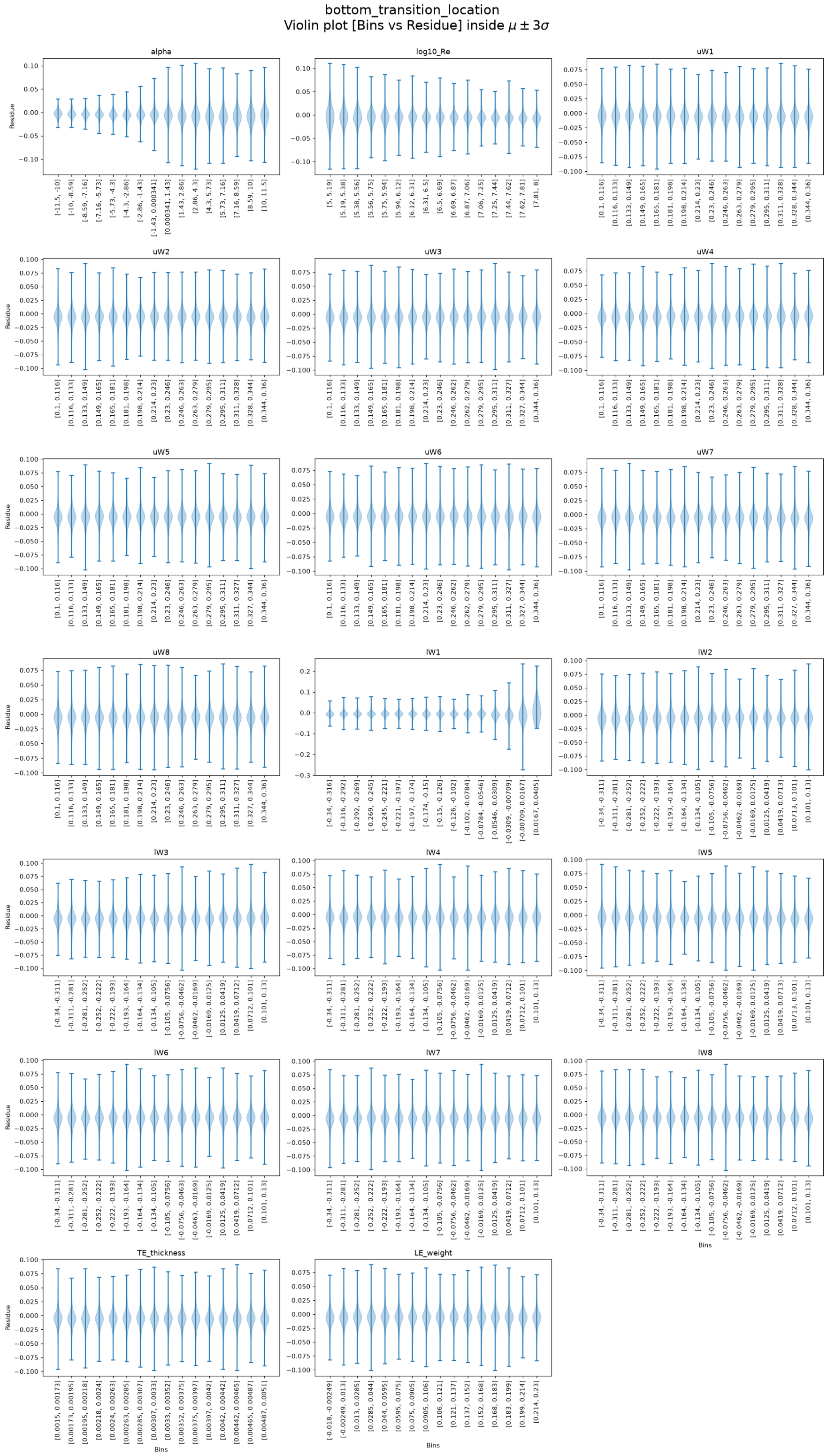
Residue — CM


CM
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$
alpha
log10_Re
uW1
uW2
uW3
uW4
uW5
uW6
uW7
uW8
lW1
lW2
lW3
lW4
lW5
lW6
lW7
lW8
TE_thickness
LE_weight
Residue
Bins

Residue — top_transition_location

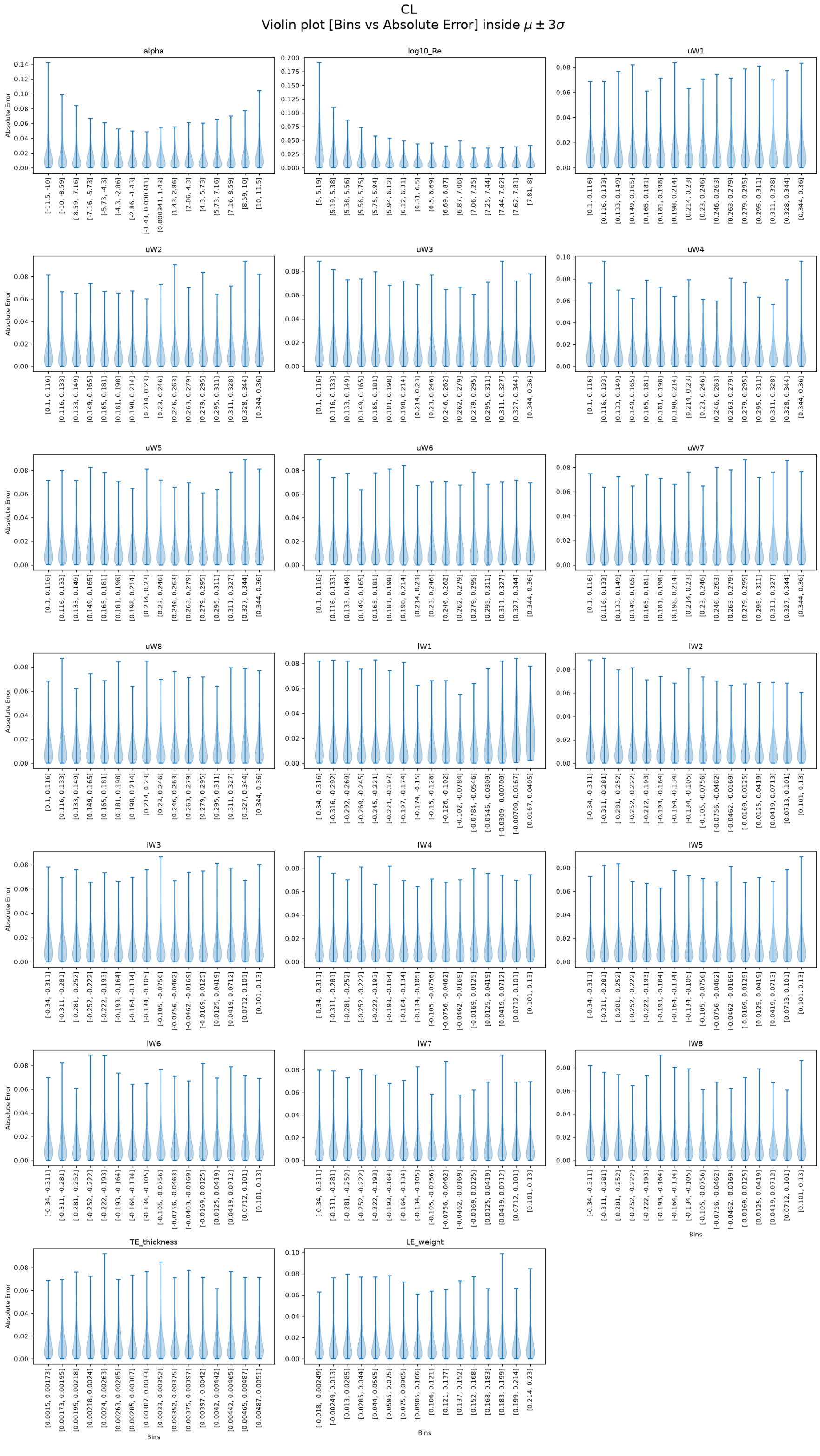


Residue — bottom_transition_location

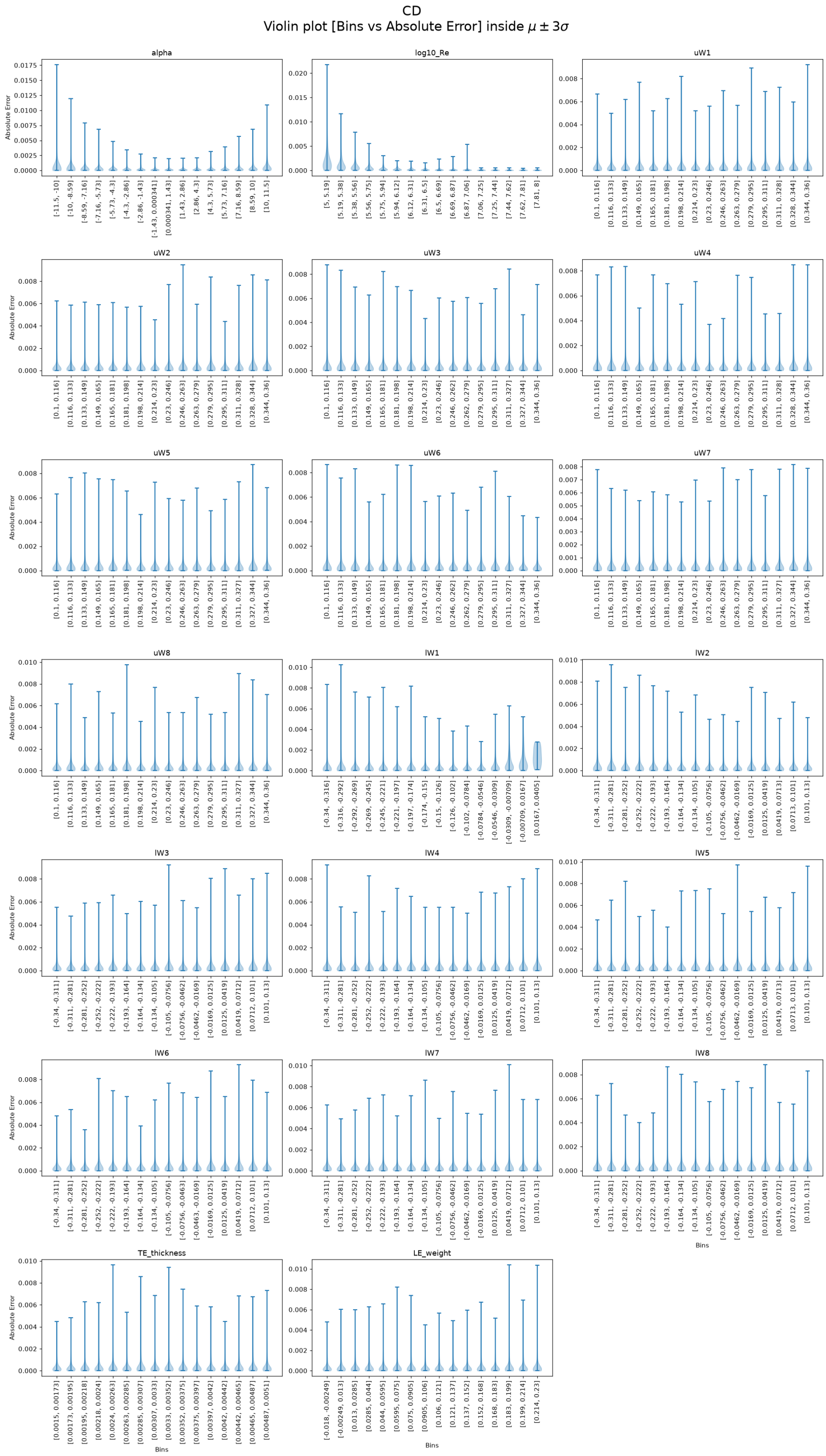


Absolute Error Violin Plots per Output (binned by input)

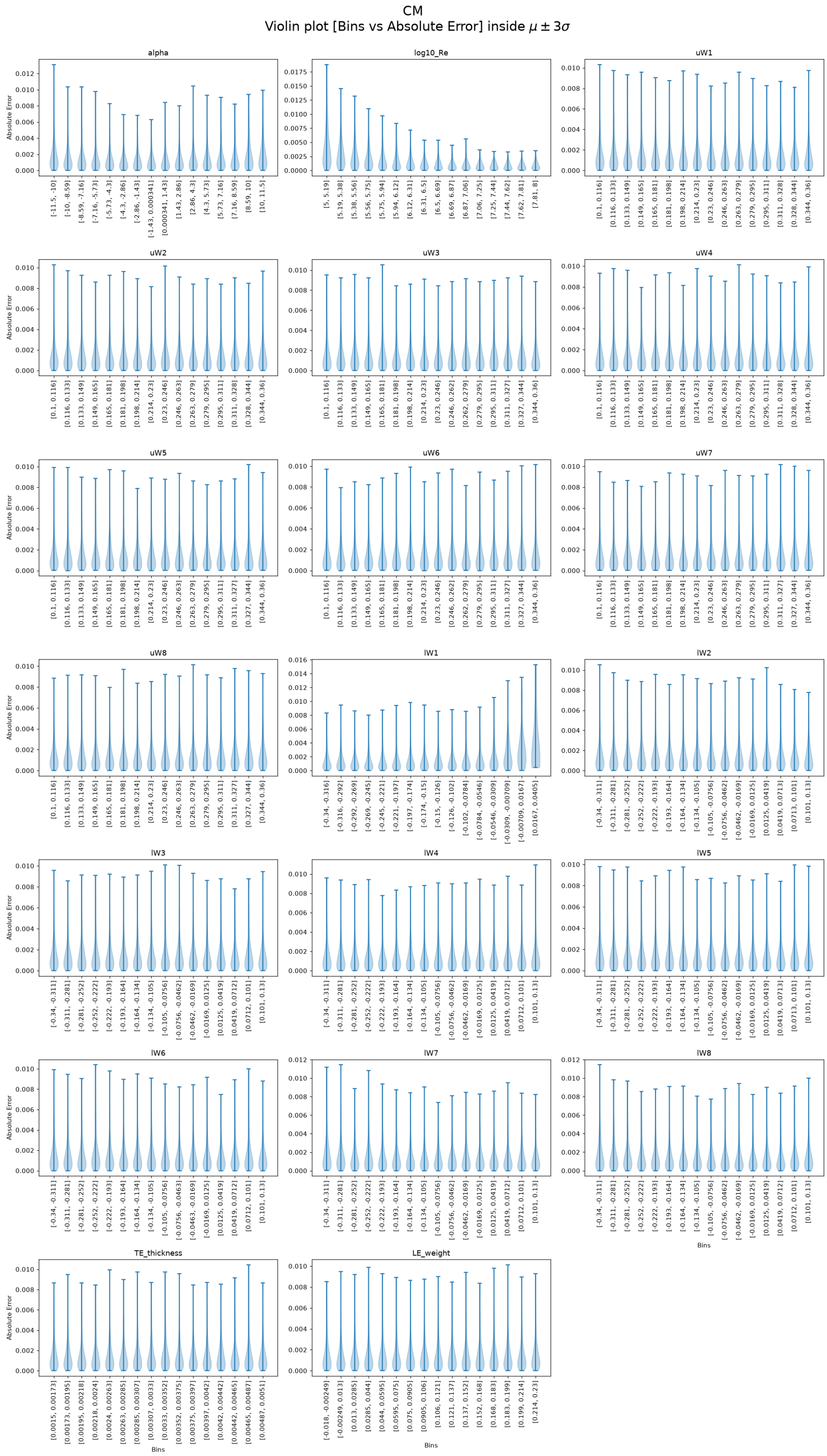
Absolute Error — CL



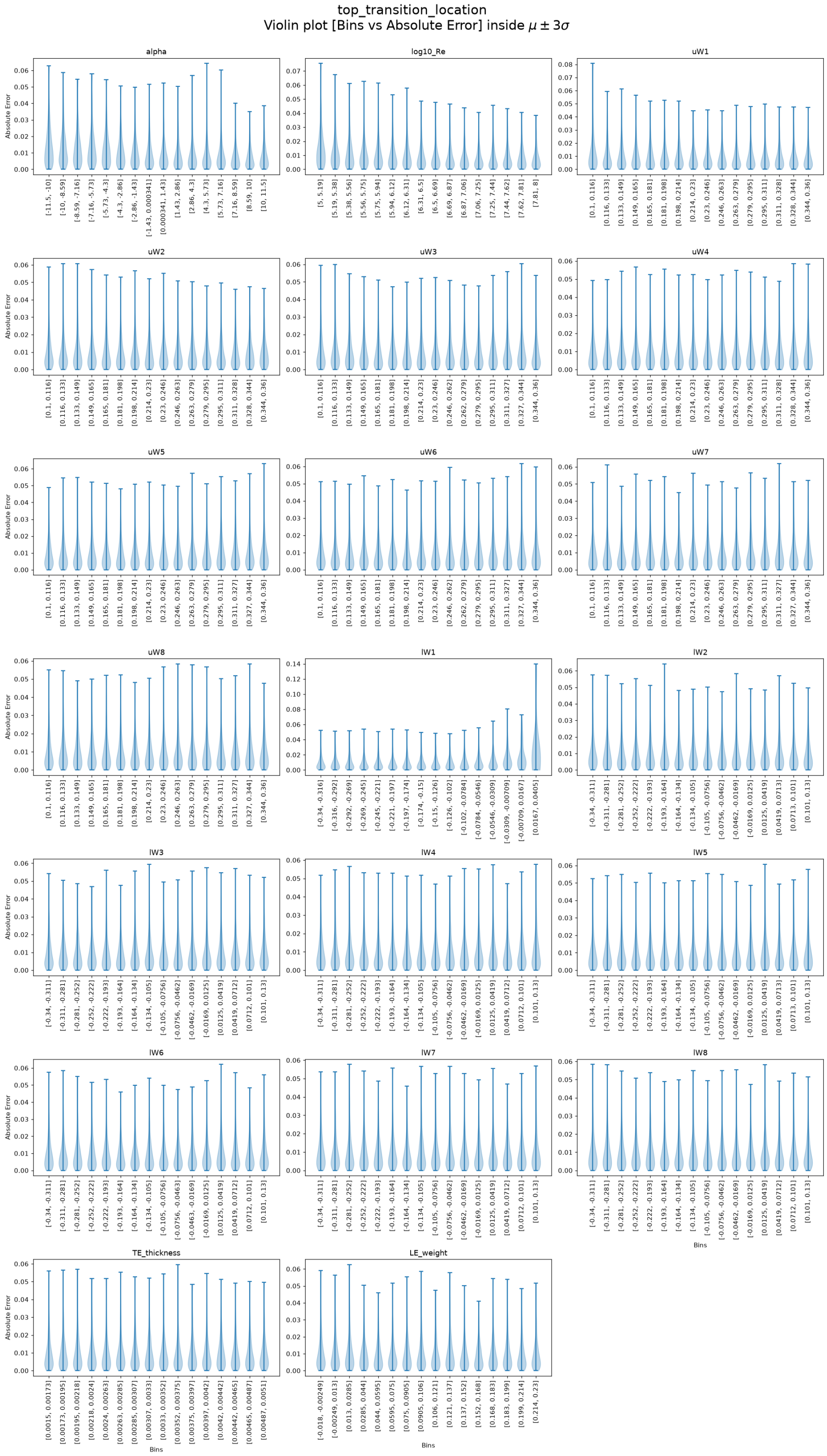
Absolute Error — CD



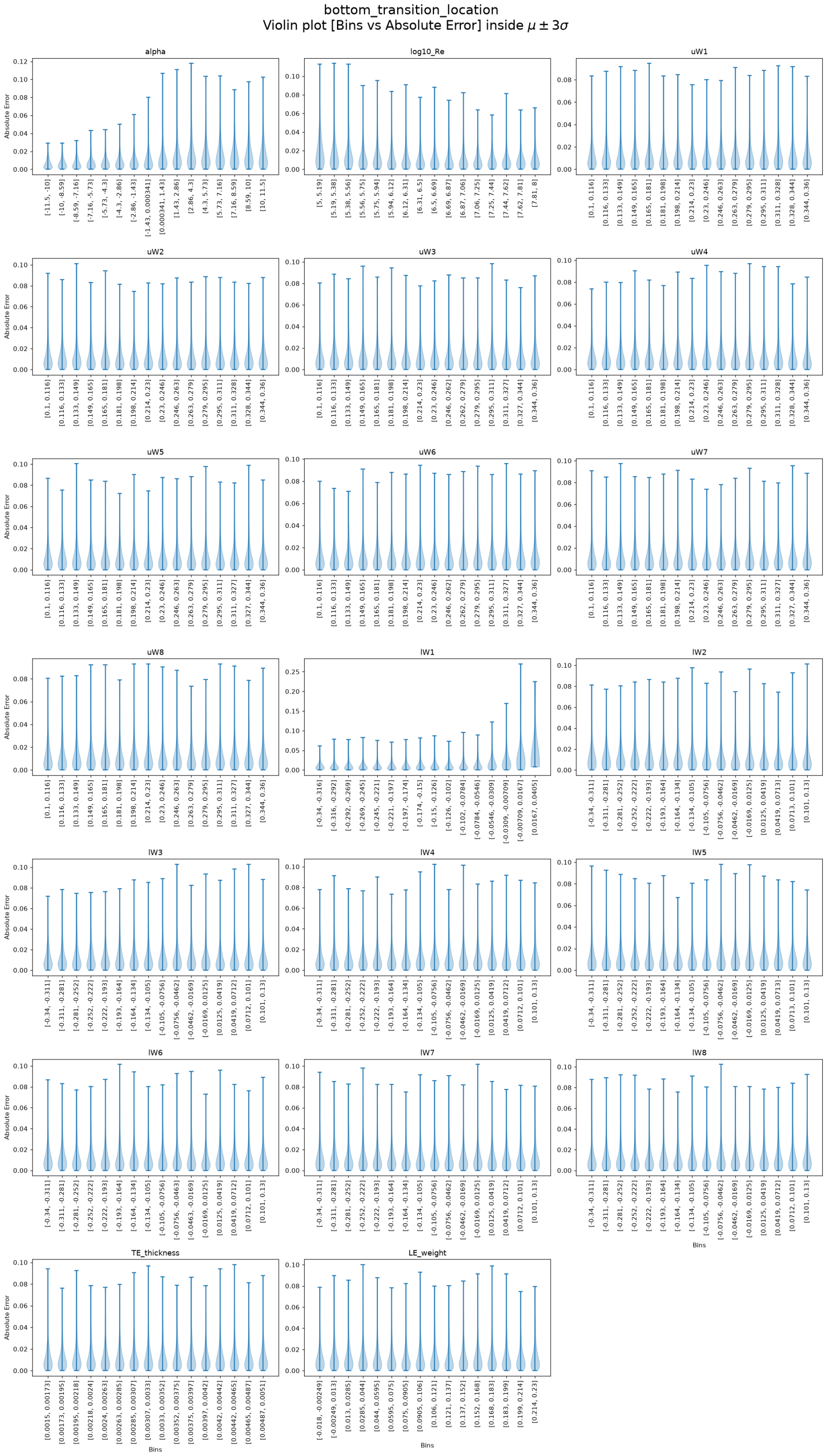
Absolute Error — CM



Absolute Error — top_transition_location



Absolute Error — bottom_transition_location

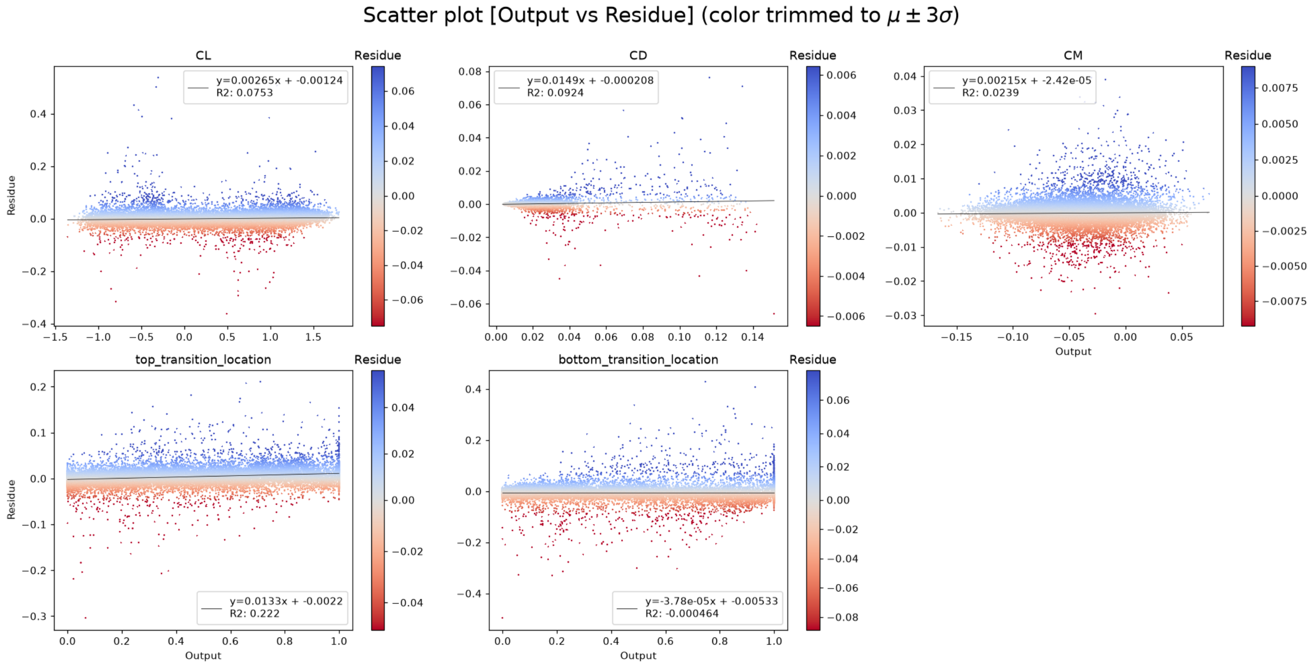


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

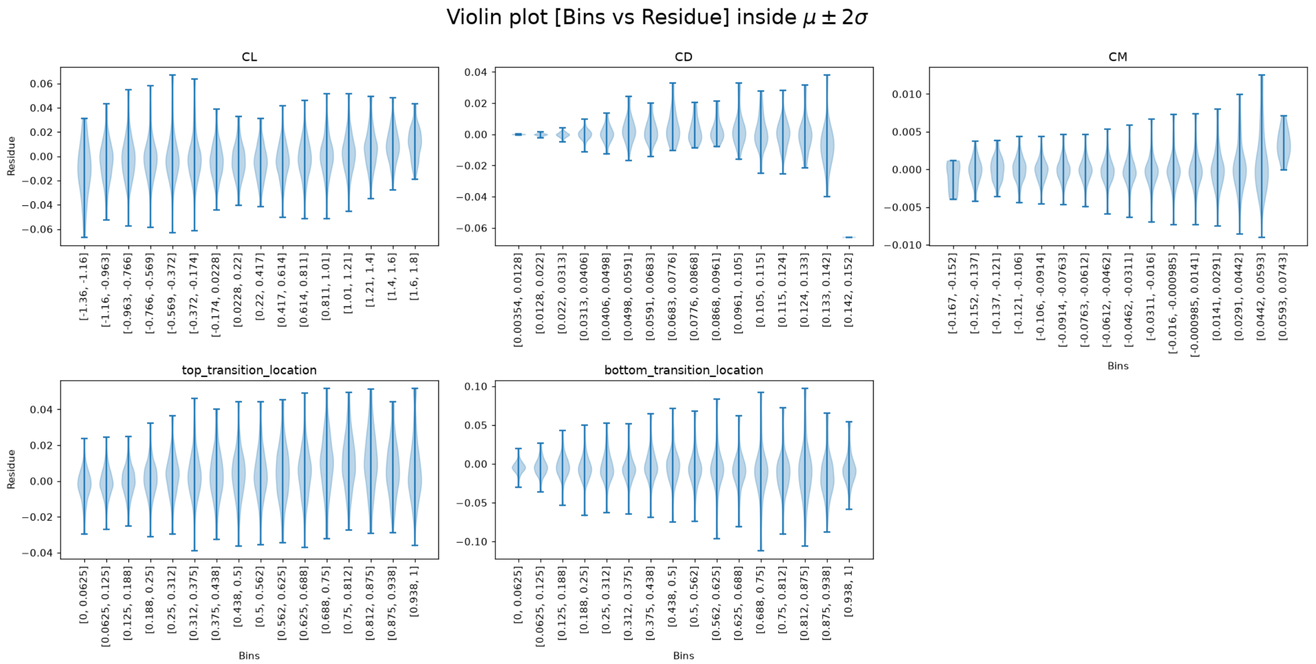
Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.



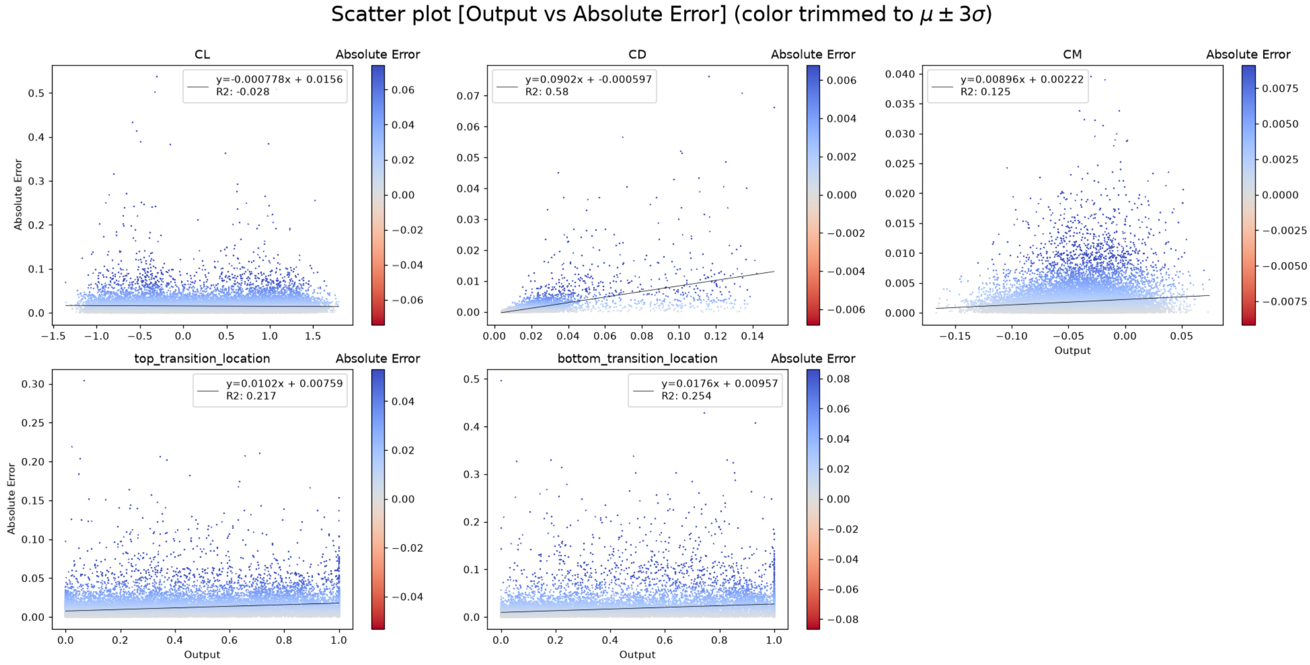
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

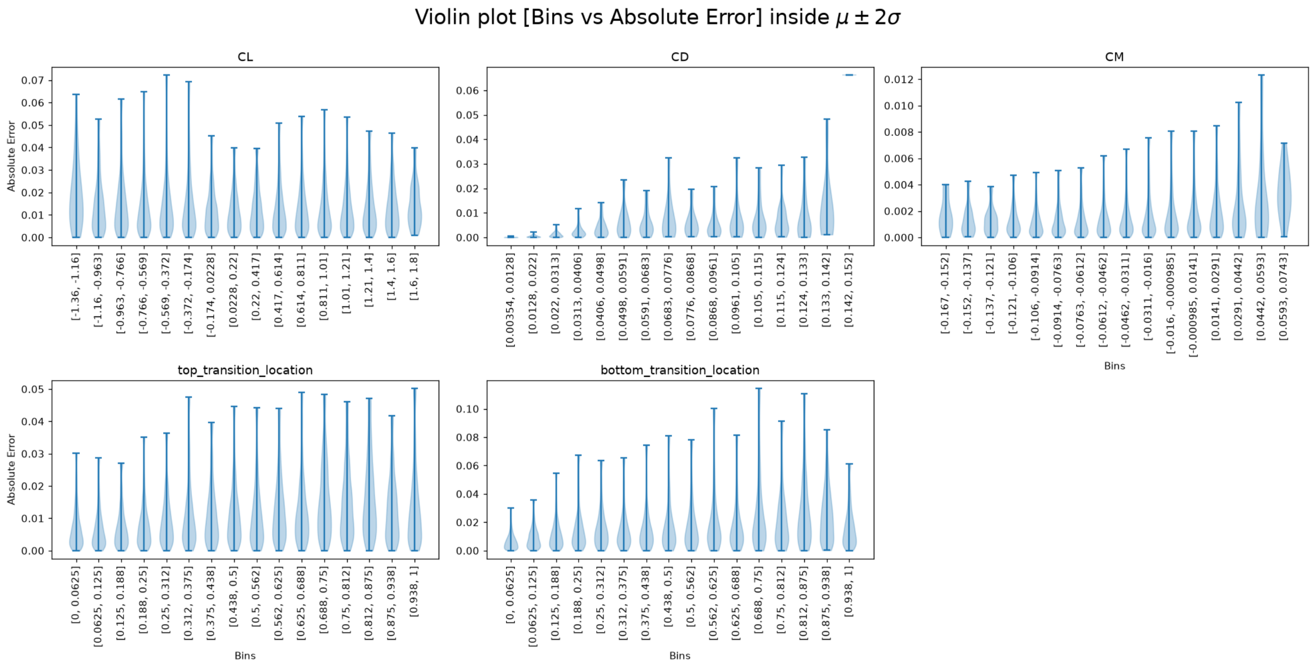


Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.



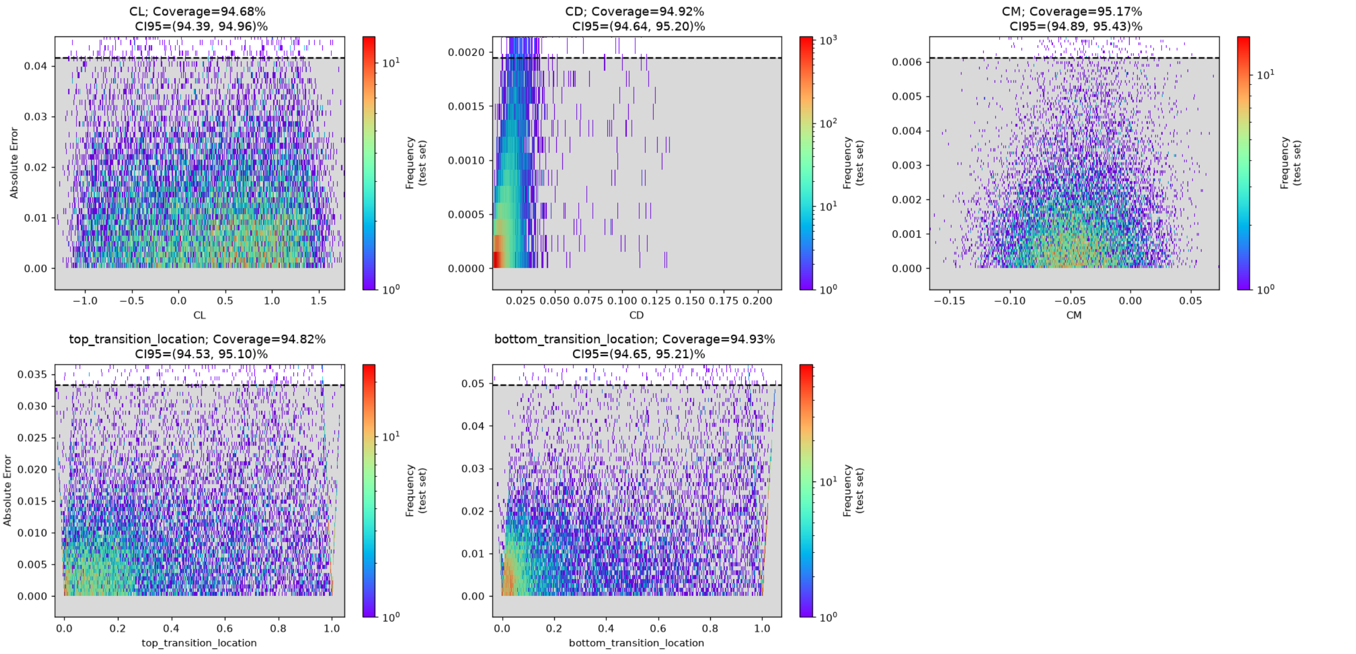
Absolute Error Violin vs True Output Bins



Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	CL	CD	CM	top_transition_location	bottom_transition_location	TOTAL COV
Coverage	94.68	94.92	95.17	94.82	94.93	—
CI	[94.39, 94.96]	[94.64, 95.2]	[94.89, 95.43]	[94.53, 95.1]	[94.65, 95.21]	—

Empirical coverage of the 95th-percentile uncertainty bound on the test split.